Calculating Pressure Changes in a Soda Can

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Final Project Date 4/23/2025

***Abstract*—The main objective of this study was to measure the change in internal pressure of a soda can before and after exposure to the atmosphere. Two methods were employed to calculate the pressure of the soda can. The first method utilized strain gauges, with two strain gauges placed orthogonally along the principal axis of the can. This method captured pressure changes associated with both the hoop stress and the longitudinal stress. Strain was then determined by measuring the voltage changes in each strain gauge. These strain values were then used to find the stress values and both the strain and stress were used to calculate each corresponding change in pressure. This method yielded two pressure change values, P_L and P_H. The second method involved measuring the diameter of the can before and after opening it. The change in diameter was then used to calculate one value for the pressure change, P_D. The values of all three calculated pressures were P_L = 405.29 ± 12.15 kPa from the longitudinal strain gauge, P_H = 499.98 ± 14.98 kPa from the hoop strain gauge, and P_D = 674.42 ± 101.05 kPa from the second method. When comparing the two methods the strain gauge method was found to be more precise after performing an uncertainty analysis for all pressures.**

***Index Terms*—Data Acquisition Device, Thin-Walled Pressure Vessel, Strain Gauge, Wheatstone Bridge**

INTRODUCTION

THE main objective of this study was to determine the pressure in a soda can using strain gauges. Two strain gauges were used with one mounted in the hoop direction and the other in the longitudinal direction. The gauges were wired to separate Wheatstone bridges (WSB) and amplifiers which were powered by a single Data Acquisition device (DAQ). The soda can was subject to different conditions and a LabVIEW virtual instrument (VI) collected data on the voltage changes measured by the DAQ. Stress and strain were then calculated for the hoop and longitudinal directions based on this data, and the pressure inside the can was solved using the data from each strain gauge. A second method of measuring the pressure in the can was also developed that involved measuring the diameter of the can before and after opening it. The results of both methods were compared with other student data and generally accepted soda can pressure values to determine the accuracy of the experiment.

Engineering stress, σ in MPa, is the intensity of internal force a material experiences at a specific point [1]. Engineering strain, ε in mm/mm, is the change in length of a measurement with respect to the initial measurement. Hooke's law, shown in (1), is used to find the modulus of elasticity, E in GPa [2]. Ductile materials will initially experience linearly elastic or non-permanent deformation with a slope of E.

$$E = \frac{\sigma}{\varepsilon} \tag{1}$$

Applying a stress that causes strain in one direction results in the material experiencing strain in a direction perpendicular to the original one. Dividing the resultant or lateral strain, ε_{lat} in mm/mm, by the original or axial strain, ε_{ax} in mm/mm, allows Poisson's ratio, ν, to be found with (2), which characterizes the relationship between these two strains.

$$\nu = -\frac{\varepsilon_{lat}}{\varepsilon_{ax}} \tag{2}$$

Thin-walled pressure vessels are cylindrical structures characterized by wall thicknesses much smaller than their diameters. This permits the assumption that thickness is constant, stress is uniformly distributed along the wall, and strain and stress normal to the surface are negligible.

Principal stresses are the maximum, σ_1 in MPa, and minimum, σ_2 in MPa, normal stresses a material experiences at a specific point. When a material is aligned with its principal directions, σ_1 and σ_2 occur simultaneously in perpendicular directions, with no shear stress present. There is also a third principal stress, σ_3 in MPa, experienced, but because stress and strain are only measured at the surface of the material and an assumed biaxial stress state, it will be zero for this study.

For thin-walled pressure vessels, internal pressure induces the principal stresses known as hoop stress, σ_H in MPa, and longitudinal stress, σ_L in MPa (Fig. 1).

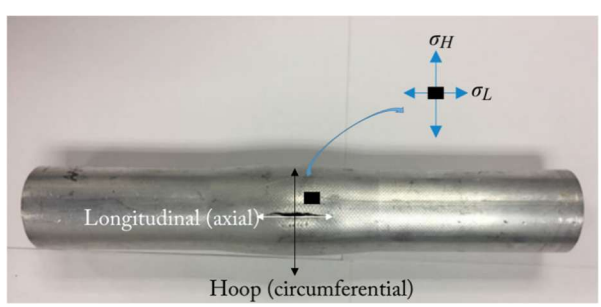

Fig. 1. Hoop and longitudinal stress on a thin-walled pressure vessel [1].

The former is equivalent to σ_1 and acts tangentially to the pressure vessel's circular cross-section. The latter is equivalent to σ_2 and acts along the pressure vessel's longitudinal axis, perpendicular to σ_H. The internal pressure within the vessel, P in kPa, vessel inner radius, r in mm, and vessel wall thickness, t in mm, can be utilized to find σ_H and σ_L with (3) and (4), respectively.

$$\sigma_H = \frac{Pr}{t} \tag{3}$$

$$\sigma_L = \frac{Pr}{2t} \tag{4}$$

Analyzing (3) and (4), σ_H is twice as large as σ_L in magnitude. However, due to human error in the strain gauge placement that causes them to not be exactly perpendicular to each other or aligned with the principal directions, as well as resolution issues from the analog-to-digital conversion associated with the DAQ, the results will not follow this relationship. Despite that, this relationship can still be used to determine how accurate the measured values for σ_H and σ_L are by using (5), where R is the ratio of hoop and longitudinal stress.

$$R = \frac{\sigma_H}{\sigma_L} \tag{5}$$

Mohr's circle, a graph that allows visualization of the different states of normal and shear stress along different planes at a specific point on a material, is useful for showing the relationship between normal and shear stress. For a thin-walled pressure vessel, σ_L is the smaller x-intercept and σ_H is the larger x-intercept (Fig. 2).

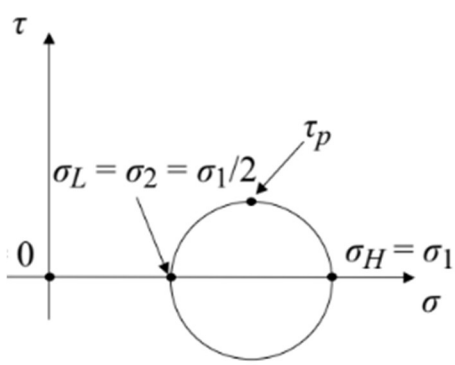

Fig. 2. Mohr's circle for a thin-walled pressure vessel with σ_L, σ_H, and in-plane maximum shear stress [1].

A strain gauge is a transducer that measures surface strain by using a thin metallic foil bonded to a material undergoing deformation. Two strain gauges were affixed to the surface of the soda can to measure the strain it experienced when subject to varying conditions. Although strain gauges are sensitive to axial, lateral, and shear strain, only axial strain is of interest here [3]. To account for the influence of lateral and shear strain on the measurements, a gauge factor, G_f, is applied.

The strain gauges were each integrated into a 1/4 WSB, amplifier, and DAQ [3]. The WSB is a circuit composed of three dummy 120 Ω resistors that is completed once the strain gauge is connected (Fig. 3) [4]. It facilitates in estimating the change in resistance produced by the strain gauge, with this resistance change corresponding to a proportional change in the voltage across the bridge, V_g in V, which is shown as V (Fig. 3). The 1/4 WSB utilized is a balanced WSB because all the resistors within it are the same. This causes the strain gauge's resistance changes to generate a voltage change that is centered around 0 volts, which allows the DAQ's analog-to-digital converter (ADC) to measure V_g effectively [5]. The amplifier is used to enlarge V_g into an amplified voltage, V_{amp} in V. It increases V_g by a factor of $gain$, the amplifier's gain value.

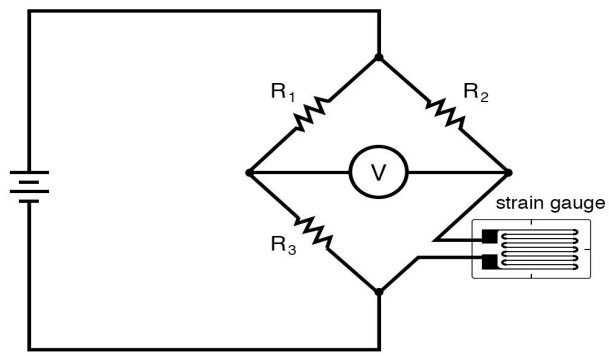

Fig. 3. 1/4 WSB circuit example with strain gauge attached [6].

A Simple Acquisition Development and Instrumentation (SADI) 2.7 DAQ was utilized with a 14-bit ADC to convert the analog voltages collected into digital values [3]. As data was recorded at a sampling rate of 1000 samples per second, the continuous voltage signals were represented as discrete values. However, with the ADC limited to only 16,383 distinct values, some accuracy was lost since voltage values could not be represented between the ADC's resolution [7]. Different sample windows are available that allow the DAQ to capture different voltage ranges. Larger sample windows allow for a broader range of data to be measured but have less precision, while smaller sample windows have more precision but limit the range of data that can be collected. The DAQ sends this voltage data to a LabVIEW VI, where it is displayed and utilized for the calculation of other values.

Strain measured by the strain gauges, ε_V in mm/mm, due to the soda can deforming from internal pressure can be found with (6) using the excitation voltage, V_s in V, V_{amp}, G_f, and $gain$.

$$\varepsilon_V = \frac{4V_{amp}}{V_s G_f \, gain} \tag{6}$$

Combining (1), (2), and (6), σ_H and σ_L can be found with (7) and (8), respectively, using E, ν, ε_{VH} in mm/mm, the strain due to hoop stress found with a strain gauge, and ε_{VL} in mm/mm, the strain due to longitudinal stress found with a strain gauge.

$$\sigma_H = \frac{E}{1+\nu^2}(\varepsilon_{VH} + \nu\varepsilon_{VL}) \quad (7)$$

$$\sigma_L = \frac{E}{1+\nu^2}(\varepsilon_{VL} + \nu\varepsilon_{VH}) \quad (8)$$

The internal pressure within the soda can due to hoop stress, P_H in kPa, and longitudinal stress, P_L in kPa, were determined with (9) and (10), respectively, using t, σ_H or σ_L, D_i in mm, the soda can's diameter before being opened, D_f in mm, the soda can's diameter after being opened, and $\sigma_{H,0}$ or $\sigma_{L,0}$ in MPa, the hoop or longitudinal stress experienced before the soda can was opened when subject to varying conditions.

$$P_H = 2t\left(\frac{\sigma_{H,0}}{D_i} - \frac{\sigma_H}{D_f}\right) \quad (9)$$

$$P_L = 4t\left(\frac{\sigma_{L,0}}{D_i} - \frac{\sigma_L}{D_f}\right) \quad (10)$$

The strain of the soda can's diameter, ε_D in mm/mm, was calculated with the measured D_i and D_f using (11).

$$\varepsilon_D = \left|\frac{D_f - D_i}{D_i}\right| \quad (11)$$

The total internal pressure within the soda can, P_D in kPa, was computed from E, t, ε_D, D_i, and v with (12).

$$P_D = \frac{2Et\varepsilon_D}{r(2-v)} \quad (12)$$

Internal pressure was found with two methods. The first utilized strain gauges and (6) – (10) to find the internal pressure generated by each stress. The second utilized diameter measurements, (11), and (12) to find the total internal pressure generated within the soda can. Uncertainty values were obtained from rules of thumb or manufacturers [3]. A t-distribution of 19 samples was utilized to find the uncertainty in the mean of the internal pressure values collected by a third party of undergraduate students who repeated the same experiment. For data where calculations were needed to determine the uncertainty values, RSS equations were utilized.

PROCEDURE

The required materials for this lab were a DAQ, two WSBs and amplifiers, laptop capable of running LabVIEW, USBC cord, strain gauges, 2 conductor strain gauge wire, solder iron and wire, clear tape, glue, sandpaper, alcohol wipes, three-inch manual micrometer, digital micrometer, wire stripper, snap knife, wood block, helping hand tweezers, and a soda can [8]. The first part of this lab procedure was to mount the strain gauges on the soda can in proper hoop and longitudinal positions. Once completed, the experimental part of the procedure involved subjecting the soda can to different conditions to measure the amplified voltages and determine the pressure of the can. Both the initial setup and experimental procedures are detailed below.

Initial Setup Procedure

The first step of the initial setup phase was to prepare the surface of the soda can for the strain gauges. A strip of sandpaper was used to remove paint and surface irregularities on the area of the can where the strain gauge would be placed [8]. The surface of the can was then cleaned with an alcohol wipe to remove the debris from the sandpaper and reveal the smooth surface. After this, the strain gauges were carefully retrieved, ensuring they did not contact human skin as the natural oils would potentially impact functionality. For this experiment, two strain gauges were retrieved as one would be oriented in the hoop direction and the other oriented in the longitudinal direction to measure ε_H and ε_L, respectively.

With the strain gauges retrieved, the mounting process was initiated. A piece of clear tape was used to pick up the first strain gauge. The strain gauge was aligned over the sanded surface in the hoop position which meant its longer side was parallel with the top and bottom of the can. Once the strain gauge was aligned correctly, the half of the tape on the side of the gauge with the leads was pressed down [8]. The other half of the tape was not pressed down so it could be lifted to apply glue under the strain gauge. The tape was carefully lifted on this side and a small dot of glue was applied directly under the strain gauge on the surface of the can. The lifted side of the tape was then pressed down and smoothened to ensure the glue was evenly applied through the area. The soda can was then set aside to give the glue time to dry.

Once enough time had passed (approximately two minutes), the clear tape was gently removed by starting at the corner and pulling in a diagonal direction aligned with the corner of the strain gauge [8]. The speed of the peeling began slow but then increased in speed when the position of the strain gage was approached. This removal method ensured the strain gauge and leads were not disturbed. Next, the longitudinal strain gauge had to be placed. Clear tape was used again to pick up the strain gauge, but this time it was aligned in the longitudinal position, orthogonal to the hoop strain gauge (Fig. 4).

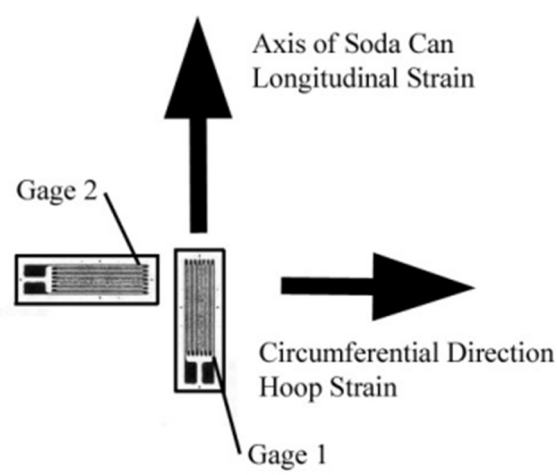

Fig. 4. Orthogonal positioning of hoop and longitudinal strain gauges [9].

The specific placement of the strain gauge did not matter as long as it remained orthogonal to the first one. In this case, it was placed above the hoop strain gauge on the sanded surface of the can. The same method of taping and glueing was utilized as before to mount the strain gauge to the can. Now that both strain gauges were mounted to the can, the next step was to connect their leads to the 2 conductor wires via soldering.

Two segments of 2 conductor strain gauge wire were retrieved [8]. Starting with one segment, a handheld wire stripper was used to remove approximately one centimeter of insulation from both ends of the wire. Next, the wire was placed on the wooden block and the blade of the snap knife was aligned at the connection point of the two wires about four centimeters from the end. The knife was then pressed down and slowly cut through the connecting region of the wires to separate them along the four-centimeter distance. The ends of the wires were then pulled apart to ensure the knife had fully cut through. This was done for both ends of the wires as it would make for easier soldering to the strain gauge leads and easier connection to the strain gauge terminals on the amplifier breadboard. The same process was repeated for the second 2 conductor wire segment.

The stripped ends of the wires had to be connected to the leads of the strain gauges by using a solder iron [8]. To aid in the process, the 2 conductor wire was secured in helping hand tweezers to be held in place during soldering. The solder iron was heated up and put in contact with solder wire to obtain molten metal at the tip. The wire and lead were brought close together and the molten metal on the solder iron was applied to the wire-lead connection point. This process was done for both leads on both strain gauges. Once the solder iron was removed, the leads of both strain gauges were connected to 2 conductor wire tips through hardened solder.

Before beginning the experiment, the diameter of the unopened can had to be measured using the three-inch manual micrometer. The micrometer was opened wide enough to fit the unopened can between the anvils and the thimble was turned until the anvils clamped down on the can. The unopened can was left in the micrometer while reading the measurements and was locked to ensure that the measurements did not change due to minor movements of the can. In order to determine the value of the 4th digit of the micrometer, the vernier values were interpolated [1]. The resultant measurement was in inches and needed to be converted to millimeters for later use.

VI Setup

The VI takes V_s and V_{amp} from each of the two strain gauges as inputs from the DAQ. The user inputs values for $V_{ampH,tare}$ and $V_{ampL,tare}$. The VI calculates the mean and standard deviation of the last 1000 samples of each V_s and V_{amp}, outputting this information. It also uses two subVIs to calculate $\varepsilon_H/\varepsilon_L$ and σ_H/σ_L. The first takes ΔV_{amp}, V_s, G_f, and $gain$ as inputs, utilizes (6), and outputs $\varepsilon_H/\varepsilon_L$. The second takes ε_H, ε_L, E, and v as inputs, utilizes (7) and (8), and outputs σ_H/σ_L. The VI also computes and outputs σ_H, σ_L, ε_H, and ε_L.

Experimental Procedure

Two 1/4 WSBs with amplifiers were wired to one DAQ to collect data for this experiment. This was done so that there were no inconsistencies between the data, as multiple DAQs would have led to data points being taken at slightly different times between the two since each DAQ captures 1000 samples per second.

To record V_s, two wires were soldered to each WSB in channels B+ and B- [3]. For the first WSB, B+ was wired to AIN0+ and B- was wired to AIN0- on the DAQ (Fig. 5). For the second WSB, B+ was wired to AIN2+ and B- was wired to AIN2-. To measure V_{amp}, two wires were connected to each WSB in channels OUTA and OUTB. For the first WSB, OUTB was wired to AIN1+ and OUTA was wired to AIN1-. For the second WSB, OUTB was wired to AIN3+ and OUTA was wired to AIN3-. The two wires used to power the WSBs were connected to channel +3.3V on the DAQ and VIN on each WSB, while the two wires used to ground the WSBs were connected to the channels labeled GND on the DAQ and GND on each WSB. Two extension wires connected each WSB to its strain gauge, with the wires being connected to the two channels labeled strain gauge on the WSB.

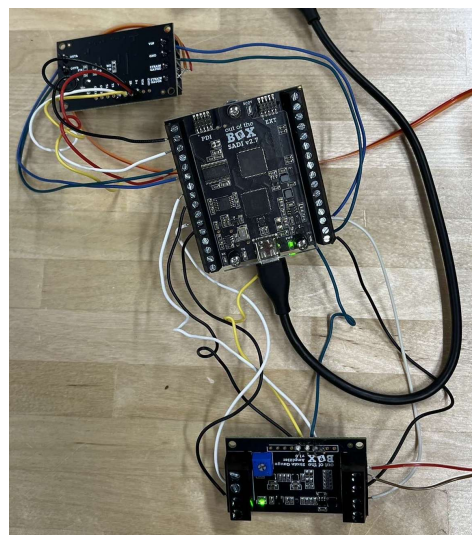

Fig. 5. One DAQ and two WSB setup utilized for this experiment.

Once the WSBs and DAQ were properly wired, the USBC cord connected to the DAQ was plugged into the laptop and the LabVIEW program was initiated. The trim potentiometer on the amplifier was adjusted until the magnitude of the mean V_{amp} was less than or equal to 0.009 volts [8]. Once this value was achieved, it was then copied and pasted into the tare control. This taring operation of V_{amp} is illustrated by equation (13) below where ΔV_{amp} is the change in amplified voltage, V_{amp} is the mean amplified voltage, and $V_{amp,tare}$ is the tare value being subtracted.

$$\Delta V_{amp} = V_{amp} - V_{amp,tare} \tag{13}$$

After taring the mean V_{amp}, the correct sample windows had to be set [8]. The ±2.56 V sample window was chosen for channel 0 and 2 since that was the smallest window that contained the value of V_s equal to approximately 2.50 V. For channels 1 and 3, the ±5.12 V sample window was chosen as a conservative estimate of what the voltage may reach. Due to the "one and done" nature of this experiment, there was no way to test the voltage change that would occur when the can was

opened beforehand. However, since the highest possible voltage reading from the DAQ power supply was 3.3 V, this value could not be exceeded and therefore ±5.12 V was selected as a safe choice.

With the amplified voltage tared and the sample windows set, the can could now be subjected to different conditions to take data. First, a data point was taken to serve as a baseline. Next, the can was squeezed in the hoop direction near the strain gauges and a data point was taken during the action. Then, a data point was taken while pressing down on the can in the longitudinal direction by applying pressure with an open palm to the top of the can. The soda can was then shaken up, being careful not to disturb the wiring, and a data point was taken shortly after. The soda can was then given adequate time to settle before opening it up. Once enough time had passed, the can was opened as the final test condition. After leaving the can open for a few seconds to let the gas out, the last data point was taken, and the program was terminated. The data was saved to an Excel file with a .xls filename extension.

The final part of the experimental procedure was to measure the opened can diameter and the wall thickness. The three-inch manual micrometer was once again used in the same fashion to measure the diameter, being extra careful not to compress the can. Then, a portion of the sanded surface of the can was removed with the snap knife by stabbing it four times in a square shape. A digital micrometer was used to measure the thickness of the sanded portion of the can (wall thickness) which the strain gauges were mounted on. All this data was recorded and saved for later analysis, and any remaining trash was disposed of.

RESULTS

The direct measurements of the wall thickness, t, unopened and undisturbed can diameter, D_i, and opened can diameter, D_f, are shown below (Table I).

TABLE I
SODA CAN GEOMETRIC DATA

Parameter	Value (mm)	Uncertainty
t	0.128	± 0.001
Di	66.03	± 0.0025
D_f	64.77	± 0.0025

The material properties of the aluminum alloy 3004 the soda can was made from are detailed below, along with G_f and $gain$ [10] (Table II). These were necessary to relate the calculated strains to stresses.

TABLE II
GIVEN MEASUREMENTS & SODA CAN MATERIAL PROPERTIES

Parameter	Units	Value	Uncertainty
E	GPa	69.0	± 0.9
G_f	None	2.085	± 0.0104
$gain$	None	900	± 4.5
v	mm/mm	0.35	±10%

The voltage tare values used in (13) are shown below where $V_{ampH,tare}$ is the tare voltage for the hoop direction in V and $V_{ampL,tare}$ is the tare voltage for the longitudinal direction in V (Table III).

TABLE III
VOLTAGE TARE VALUES

Symbol	Value (V)	Uncertainty (V)
$V_{ampH,tare}$	0.0042	± 0.0056
$V_{ampL,tare}$	0.0112	± 0.0148

All voltages corresponding to the longitudinal strain gauge are detailed in Table IV below. These include the excitation voltage of the WSB connected to the longitudinal strain gauge, $V_{s,L}$, the voltage produced from the undisturbed can condition, $V_{ampL,0}$, the voltage produced after squeezing the can, $V_{ampL,1}$, the voltage produced after pressing down on the can, $V_{ampL,2}$, the voltage produced after shaking the can, $V_{ampL,3}$, and the voltage produced after opening the can, $V_{ampL,4}$.

TABLE IV
VOLTAGE DATA FROM LONGITUDINAL GAUGE

Parameter	Value (V)	Uncertainty (V)
$V_{s,L}$	2.5004	±0.0002
$\Delta V_{ampL,0}$	0.0250	±0.0084
$\Delta V_{ampL,1}$	-0.0501	±0.0423
$\Delta V_{ampL,2}$	-0.0793	±0.0146
$\Delta V_{ampL,3}$	0.0296	±0.0079
$\Delta V_{ampL,4}$	-0.0934	±0.0082

The voltages detailed in Table V below are the same as those listed above, but now for the hoop strain gauge rather than the longitudinal one.

TABLE V
VOLTAGE DATA FROM HOOP GAUGE

Parameter	Value (V)	Uncertainty
$V_{s,H}$	2.5006	±0.0002
$\Delta V_{ampH,0}$	0.0092	±0.0079
$\Delta V_{ampH,1}$	0.2705	±0.1803
$\Delta V_{ampH,2}$	0.0742	±0.0106
$\Delta V_{ampH,3}$	0.0197	±0.0081
$\Delta V_{ampH,4}$	-1.8300	±0.0079

The LabVIEW program also calculated the strain values for all five conditions from both the longitudinal and hoop strain gauges using (6). The strain values from the longitudinal strain gauge, $\varepsilon_{V,L}$, are presented in Table VI below and the strain values from the hoop strain gauge, $\varepsilon_{V,H}$, are presented in Table VII below. An additional strain from the change in can diameter before and after opening, ε_D, was calculated to be equal to 0.0191 ± 0.0092 mm/mm using (11).

TABLE VI
STRAIN DATA FROM LONGITUDINAL GAUGE

Parameter	Value (mm/mm)	Uncertainty (mm/mm)
$\varepsilon_{V,L0}$	0	± 0
$\varepsilon_{V,L1}$	-0.63×10^{-4}	$\pm 5.21 \times 10^{-7}$
$\varepsilon_{V,L2}$	-0.89×10^{-4}	$\pm 7.43 \times 10^{-7}$
$\varepsilon_{V,L3}$	-0.04×10^{-4}	$\pm 3.19 \times 10^{-8}$
$\varepsilon_{V,L4}$	-1.01×10^{-4}	$\pm 8.32 \times 10^{-7}$

TABLE VII
STRAIN DATA FROM HOOP GAUGE

Parameter	Value (mm/mm)	Uncertainty (mm/mm)
$\varepsilon_{V,H0}$	-0.02×10^{-4}	$\pm 1.64 \times 10^{-8}$
$\varepsilon_{V,H1}$	2.21×10^{-4}	$\pm 1.72 \times 10^{-6}$
$\varepsilon_{V,H2}$	0.54×10^{-4}	$\pm 4.35 \times 10^{-7}$
$\varepsilon_{V,H3}$	0.07×10^{-4}	$\pm 5.66 \times 10^{-8}$
$\varepsilon_{V,H4}$	-15.69×10^{-4}	$\pm 1.28 \times 10^{-5}$

DISCUSSION

Restatement of Objectives

The main objective of this study was to determine the pressure in a soda can through two different methods. For the first method, one strain gauge was mounted horizontally in the hoop direction, and the other was mounted vertically in the longitudinal direction. Stresses and strains were then acquired in these two directions and used to calculate the pressure inside the can. For the second method, the diameter of the can was measured before and after opening to determine the pressure. Both methods had their results compared to other students performing the test in order to determine the accuracy. The final results were compared to a generally accepted value from an online source.

Analysis of Results

As previously stated, strains were calculated using (6) from the voltage data. These strain values were then used in (7) and (8) to calculate hoop stress and longitudinal stress, respectively. Table VIII shows the calculated longitudinal stress differentials relative to the unopened and undisturbed can. $\sigma_{L,n}$ is the stress derived from the longitudinal gauge, where n is a number from zero to four used to signify the stress state the soda can was under when a data point was recorded. The subscript *n* being 0, 1, 2, 3, and 4 means the can was in the state of being undisturbed, squeezed, pressed down, shaken, and opened, respectively. Table IX shows the same data but for the strain gauge measuring hoop stresses.

TABLE VIII
STRESS DIFFERENTIALS FROM LONGITUDINAL GAUGE

Parameter	Value (MPa)	Uncertainty (MPa)
$\sigma_{L,0}$	-0.055	± 0.001
$\sigma_{L,1}$	1.183	± 0.019
$\sigma_{L,2}$	-5.457	± 0.086
$\sigma_{L,3}$	0.562	± 0.009
$\sigma_{L,4}$	-51.068	± 0.826

TABLE IX
STRESS DIFFERENTIALS FROM HOOP GAUGE

Parameter	Value (MPa)	Uncertainty (MPa)
$\sigma_{H,0}$	-0.157	± 0.004
$\sigma_{H,1}$	15.801	± 0.262
$\sigma_{H,2}$	1.954	± 0.032
$\sigma_{H,3}$	0.818	± 0.014
$\sigma_{H,4}$	-125.997	± 2.012

Effects of Squeezing the Can

When the soda can was squeezed, the calculated longitudinal and hoop stress differentials from the nominal state were 1.183 ± 0.019 MPa and 15.801 ± 0.262 MPa, respectively. The increase in stress was expected as decreasing the volume of the can would increase the internal pressure and therefore the overall stress experienced by the can. The change in hoop stress was substantially larger than the change in longitudinal stress, likely due to the way the can was squeezed. The can was squeezed by applying two fingers, one near the gauges and one on the opposite side of the can. This applied a force in the hoop direction, so it makes sense for the resultant stress to be greater in that direction. Additionally, it is possible that pressure was applied closer to the hoop gauge than the longitudinal, which would cause the hoop gauge to have a larger increase in strain. This would produce a larger stress change similar to what was calculated.

Effects of Pressing the Can

The next condition applied to the can was pressing down on the top of it. Unlike when the can was squeezed, the force was now being applied in the longitudinal direction. The resultant stress change in the longitudinal and hoop directions were -5.457 ± 0.086 MPa and 1.954 ± 0.032 MPa, respectively. These results are expected as the can was compressed in the longitudinal direction and therefore a larger negative stress will occur in that direction. On the other hand, a smaller tensile stress was experienced in the hoop direction because the can resisted the compressive axial force by expanding radially outward. This outward expansion in the hoop direction created a tensile stress at the cylindrical surface that is lower in magnitude than the compressive stress as shown in the results.

Effects of Shaking the Can

After squeezing and pressing down on the can, it was then carefully shaken up. The resultant stress change in the longitudinal and hoop directions from shaking the can were equal to 0.562 ± 0.009 MPa and 0.818 ± 0.014 MPa, respectively. Both changes in stress were positive because the pressure in the can expanded outward after shaking it. The stress increased in the hoop direction more than the longitudinal because when the can was shaken the liquid inside sloshed around causing local pressure spikes against the walls of the can which temporarily increased hoop stress. The stress in the longitudinal direction changed less dramatically because cylindrical shapes resist hoop stress less efficiently than longitudinal stress. Compared to the previous conditions, relatively small changes in both longitudinal and hoop stresses occurred. This was partially due to the soda can not being shaken hard enough or long enough in order to preserve the wiring. However, the small changes in stress were more likely caused by the stiffness of the can resisting deformation, the incompressibility of the fluid inside, and the built in air gap within the can that absorbed some of the pressure change.

Effects of Opening the Can

The final condition applied to the can was to open it up. The resultant stress change in the longitudinal and hoop directions were -51.068 ± 0.826 MPa and -125.997 ± 2.012 MPa, respectively. Unlike when the can was shaken, these stress changes were compressive (negative) and very high in magnitude compared to the previous conditions. This was expected as the internal pressure of the can would have

decreased to equalize with the surrounding atmospheric pressure. The hoop stress more drastically decreased because it was mainly caused by the internal pressure. Once the pressure was released, the can walls did not have to hold in that force anymore and therefore compressed inward.

The pressure and stress of a thin-walled pressure vessel are directly proportional as seen in (5). The ratio of hoop stress to longitudinal stress was found by dividing $\sigma_{H,4}$ by $\sigma_{L,4}$ which produced the value 2.467 ± 0.056. This differs from the expected ratio of 2 predicted by the pressure vessel theory used for the strain gauge approach, which is likely due to the strain gauges not being exactly perpendicular to each other and not exactly oriented along the longitudinal and circumferential directions. These gauge positioning errors could have propagated to produce this deviation from the theoretical hoop stress to longitudinal stress ratio.

Calculating the Internal Pressure of the Can

The internal pressure of the soda can was calculated using two methods. Method one utilized strain gauges to calculate pressure from two different strain gauges. One strain gauge was oriented in the longitudinal direction and (10) was used to calculate the pressure, P_L, from its data. Similarly, the other strain gauge was oriented in the hoop direction and (9) was used to calculate the pressure, P_H, from its data. The second method utilized the initial and final diameter of the can, corresponding to before and after it was opened, to calculate pressure, P_D, using (12). All three final pressure values are presented in Table X below.

TABLE X
CALCULATED GAGE PRESSURES

Parameter	Value (kPa)	Uncertainty (kPa)
P_L	405.29	± 12.15
P_H	499.98	± 14.98
P_D	674.42	± 101.05

The result from the second method using the diameters generated the highest pressure value and the greatest uncertainty. This suggests that this method was less precise than the strain gauge method whose results for hoop and longitudinal pressure were relatively similar.

Comparison with Provided Pressure Data

A third party comprised of undergraduate students conducted a similar experiment to find the internal gage pressure of an unopened soda can. This data set included the calculated pressures of 22 different cans through a form of the strain gauge method. The exact procedure used to acquire this data was unknown. A 95% confidence interval was conducted in order to remove outliers, which resulted in a mean of 238.365 ± 255.085 kPa. There were no outliers based on this approach. Internal gauge pressure values of 3395 Pa, 142.6 Pa, and 335 Pa were removed from the data set as they were several orders of magnitude smaller than the mean. The remaining data set is shown in Table XI. A t-distribution was then performed on the new data set using a t-value of 2.101, which resulted in a mean of 267 ± 43 kPa.

TABLE XI
PROVIDED SODA CAN PRESSURE DATA

Parameter	Value (kPa)
1	438.19
2	158.35
3	290.00
4	283.17
5	137.78
6	209.98
7	314.91
8	232.73
9	243.77
10	411.12
11	162.80
12	314.68
13	344.58
14	196.96
15	354.69
16	207.71
17	265.3
18	144.79
19	361.71

Comparing the provided pressure data mean and 95% confidence interval with the P_L, P_H, and P_D values and confidence intervals revealed there was overlap. The pressure calculated from the longitudinal gauge was the closest to the mean of the provided data, with a minimum difference of 83.14 kPa considering the true pressure values lie at the edge of the uncertainty margins. This can be a result of considerable human error in the experimental procedure detailed in this report but also in the procedure used by the undergraduate students to produce the provided pressure data. Although the students also utilized a strain gauge approach to calculate the pressure, the provided data may be unreliable because the exact process used is unknown and the 95% confidence interval of the initial data set was larger than the mean.

Comparison with Established Pressure

An online source was used to determine an established internal pressure of a 355 mL Coca-Cola can at 24° C of 380 kPa [11]. Although this value is closer to P_L, P_H, and P_D than the mean of the provided data set, it does not fall in the uncertainty window. This suggests that both methods used to calculate the gauge pressure, through strain gauges and diametral measurements, are inaccurate and unreliable. Additionally, the diametral difference method seems to largely overestimate the internal pressure. This method is highly dependent on the accuracy in which the diameters were measured. It is possible that the diameter measured after opening the can was smaller than the true value, because there was less resistance to the force applied by the micrometer due to the internal pressure drop associated with opening the can. This would have resulted in a larger change in diameter which would produce a larger calculated pressure according to (12).

Limitations of this Study

The accuracy of this experiment was limited by human and instrumental error. First, it was difficult to mount the two strain gauges completely orthogonal to one another. Even if the second gauge was aligned orthogonally to the first before pressing the tape down, the position was slightly shifted after

applying glue due to the difficult mounting process on the curved surface. Therefore, an angle not equal to 90 degrees existed between the two strain gauges. As a result, the calculated stresses were not pure hoop and longitudinal stress as each one had a small component of the other contributing to the value. This inaccuracy propagated into the pressure calculations causing the final results to be less accurate.

Another limitation affecting the accuracy of the results was present in the second method to calculate pressure using the can diameter before and after opening. Due to the ductility and thinness of the can surface, it was possible to cause a shallow dent when measuring with the micrometer if enough force was applied. This was especially true for the empty can which had less pressure inside to help keep its shape. As a result, the diameter measurements from the micrometer were slightly inaccurate which then led to propagated error in the pressure calculations. Another issue with the diameter measurements was that a random start and end point were chosen for the closed and opened measurements. Therefore, different surface imperfections at these locations could have impacted the difference between the two measurements.

As previously stated in the procedure, a conservative sample window of ±5.12 volts was chosen for the channels corresponding to the amplified voltages. This was done to ensure the voltage change would not cause saturation when the can was opened. As a result, the ADC had a lower resolution causing the voltage measurements from the DAQ to be less precise. After the experiment, it was observed that the change in voltages did not exceed ±2.56 volts and therefore that sample window could have been selected to increase resolution and precision. Finally, thermal strain could have occurred from the strain gauge resistors heating up if the experiment had a long duration [8]. This additional strain could have impacted results by increasing the stress measured in both directions.

Suggestions for Improvement

The previously stated limitations and issues could be improved upon through various experimental changes. First off, the strain gauge placement issue could be improved by using the nutrition label lines on the soda can to align the strain gauges orthogonal to each other. However, this might still prove difficult as sanding near the label may obstruct the straight lines. Therefore, another method could be to add straight line marks on the sanded surface to align the strain gauges with. This could be done using a marker and ruler or the snap knife blade. Next, the soda can diameter accuracy could be improved by measuring it multiple times and taking the mean value. An outlier analysis would then be performed using the sample size t number and the standard deviation of the measurements to find upper and lower limits. Any measurement exceeding these limits after applying the confidence interval would be removed as outliers. Additionally, the closed and opened soda can diameter measurements could be taken from the same start and end locations on the can to maximize the accuracy of these readings.

To choose the sample window to maximize precision, which in this case was ±2.56 volts rather than the ±5.12 volts used, the voltage change caused by opening the can could be approximated using manual solving methods. Research from multiple sources and more than one trial of calculations would need to be made to ensure the sample window chosen does not cause saturation. Finally, the effects of thermal strain could be better negated by performing the experiment in a cooler setting and by taking shorter breaks between applying different conditions to the can to ensure the experiment does not last too long.

Conclusion

This study successfully measured the change in internal pressure of a soda can by employing two different experimental methods: strain gauge analysis and diameter change measurement. The strain gauge method, which involved capturing both hoop and longitudinal stresses, found that P_H was equal to 499.98 ± 14.98 kPa and P_L was equal to 405.29 ± 12.15 kPa. These had relatively low uncertainties, while the diameter measurement method found that P_D was equal to 674.42 ± 101.05 kPa, which has a much larger uncertainty. This indicates that the strain gauge method offered greater precision in determining internal pressure changes within the soda can. Overall, these results highlight the effectiveness and reliability of strain gauges in accurately assessing pressure variations in thin-walled pressure vessels.

Appendix

All parameters used in this experiment had an associated uncertainty [8]. Three different methods of determining uncertainty were utilized depending on how the parameter was measured or calculated. First, some uncertainties were provided by manufacturers or known by rule of thumb (Table XII). Statistical uncertainties with a 95% confidence interval were found for the excitation voltages and tared voltage values based off 1000 samples and for the mean pressure based off the student data with a sample size of 19 and t value of 2.101 (Table XIII). Finally, the uncertainties for parameters dependent on other variables were found using RSS propagation equations that considered all contributing uncertainties (Table XIV).

TABLE XII
GIVEN, MANUFACTURER, OR RULE OF THUMB UNCERTAINTY VALUES

Variable	μ	Uncertainty	Source	Units
E	μ_E	± 0.9	Rule of Thumb [12]	GPa
ν	μ_ν	± 0.035	Manufacturer [12]	mm/mm
t	μ_t	± 0.001	Manufacturer [1]	mm
G_f	μ_{G_f}	± 0.0104	Manufacturer [1]	None
D_i	μ_{D0}	± 0.0025	Manufacturer [12]	mm
D_f	μ_{DOp}	± 0.0025	Manufacturer [12]	mm
$gain$	μ_{gain}	± 4.5	Manufacturer [1]	V/V

TABLE XIII
STATISTICAL UNCERTAINTY VALUES

Variable	μ	Uncertainty	N	Units
$V_{s,L}$	$\mu_{V_{s,L}}$	± 0.0002	1000	V
$V_{s,H}$	$\mu_{V_{s,H}}$	± 0.0002	1000	V
$V_{ampL,tare}$	$\mu_{V_{ampL,tare}}$	± 0.0148	1000	V
$V_{ampH,tare}$	$\mu_{V_{ampH,tare}}$	± 0.0056	1000	V
P	μ_P	± 43	19	kPa

TABLE XIV
RSS PROPAGATED UNCERTAINTY VALUES

Variable	μ	Uncertainty	Units
ΔV_{ampL}	$\mu_{\Delta V_{ampL}}$	± 0.0163	V
ΔV_{ampH}	$\mu_{\Delta V_{ampH}}$	± 0.0429	V
ε_V	μ_{ε_V}	± 1.72 x 10^{-6}	mm/mm
ε_D	μ_{ε_D}	± 0.0092	mm/mm
σ_L	μ_{σ_L}	± 0.2350	MPa
σ_H	μ_{σ_H}	± 0.5810	MPa
R	μ_R	± 0.056	MPa/MPa
P_L	μ_{P_L}	± 12.15	kPa
P_H	μ_{P_H}	± 14.98	kPa
P_D	μ_{P_D}	± 101.05	kPa

The uncertainty for every ΔV_{ampL}, ΔV_{ampH}, ε_V, σ_L, and σ_H measurement was calculated with the respective RSS propagation equations separately and reported in the results and discussion. The uncertainties of these parameters displayed in the table above are an average of all the separate uncertainties, as the values varied.

RSS Uncertainty Equations

RSS Propagation of (13) for ΔV_{amp}

$$\mu_{\Delta V_{amp}} = [\frac{\partial \Delta V_{amp}^2}{\partial V_{amp}}\mu_{Vamp}^2 + \frac{\partial \Delta V_{amp}^2}{\partial V_{amp,tare}}\mu_{Vamp,tare}^2]^{1/2} \quad (14)$$

RSS Propagation of (6) for ε_V

$$\mu_{\varepsilon_V} = [\frac{\partial {\varepsilon_V}^2}{\partial \Delta V_{amp}}\mu_{\Delta V_{amp}}^2 + \frac{\partial {\varepsilon_V}^2}{\partial V_s}\mu_{V_s}^2 + \frac{\partial {\varepsilon_V}^2}{\partial G_f}\mu_{G_f}^2 + \frac{\partial {\varepsilon_V}^2}{\partial gain}\mu_{gain}^2]^{1/2} \quad (15)$$

RSS Propagation of (11) for ε_D

$$\mu_{\varepsilon_D} = [\frac{\partial {\varepsilon_D}^2}{\partial D_i}\mu_{D_i}^2 + \frac{\partial {\varepsilon_D}^2}{\partial D_f}\mu_{D_f}^2]^{1/2} \quad (16)$$

RSS Propagation of (8) for σ_L

$$\mu_{\sigma_L} = [\frac{\partial {\sigma_L}^2}{\partial E}\mu_E^2 + \frac{\partial {\sigma_L}^2}{\partial \nu}\mu_\nu^2 + \frac{\partial {\sigma_L}^2}{\partial \varepsilon_{VH}}\mu_{\varepsilon_{VH}}^2 + \frac{\partial {\sigma_L}^2}{\partial \varepsilon_{VL}}\mu_{\varepsilon_{VL}}^2]^{1/2} \quad (17)$$

RSS Propagation of (7) for σ_H

$$\mu_{\sigma_H} = [\frac{\partial {\sigma_H}^2}{\partial E}\mu_E^2 + \frac{\partial {\sigma_H}^2}{\partial \nu}\mu_\nu^2 + \frac{\partial {\sigma_H}^2}{\partial \varepsilon_{VL}}\mu_{\varepsilon_{VL}}^2 + \frac{\partial {\sigma_H}^2}{\partial \varepsilon_{VH}}\mu_{\varepsilon_{VH}}^2]^{1/2} \quad (18)$$

RSS Propagation of (10) for P_L

$$\mu_{P_L} = [\frac{\partial {P_L}^2}{\partial \sigma_L}\mu_{\sigma_L}^2 + \frac{\partial {P_L}^2}{\partial \sigma_{L,0}}\mu_{\sigma_{L,0}}^2 + \frac{\partial {P_L}^2}{\partial t}\mu_t^2 + \frac{\partial {P_L}^2}{\partial D_i}\mu_{D_i}^2 + \frac{\partial {P_L}^2}{\partial D_f}\mu_{D_f}^2]^{1/2} \quad (19)$$

RSS Propagation of (9) for P_H

$$\mu_{P_H} = [\frac{\partial {P_H}^2}{\partial \sigma_H}\mu_{\sigma_H}^2 + \frac{\partial {P_H}^2}{\partial \sigma_{H,0}}\mu_{\sigma_{H,0}}^2 + \frac{\partial {P_H}^2}{\partial t}\mu_t^2 + \frac{\partial {P_H}^2}{\partial D_i}\mu_{D_i}^2 + \frac{\partial {P_H}^2}{\partial D_f}\mu_{D_f}^2]^{1/2} \quad (20)$$

RSS Propagation of (12) for P_D

$$\mu_{P_D} = [\frac{\partial {P_D}^2}{\partial E}\mu_E^2 + \frac{\partial {P_D}^2}{\partial t}\mu_t^2 + \frac{\partial {P_D}^2}{\partial \varepsilon_D}\mu_{\varepsilon_D}^2 + \frac{\partial {P_D}^2}{\partial \mathrm{r}}\mu_{\mathrm{r}}^2 + \frac{\partial {P_D}^2}{\partial \nu}\mu_\nu^2]^{1/2} \quad (21)$$

RSS Propagation of (5) for P_D

$$\mu_R = [\frac{\partial R^2}{\partial \sigma_H}\mu_{\sigma_H}^2 + \frac{\partial R^2}{\partial \sigma_L}\mu_{\sigma_H}^2]^{1/2} \quad (22)$$

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